

# Eddy Hu

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## EDUCATION

**New York University, Tandon School of Engineering** Expected May 2027  
*B.S. in Mathematics and Computer Science; GPA: 3.5/4.0*  
**Coursework:** Algorithms, Operating Systems, Software Engineering, Database Systems, Probability and Statistics, Real Analysis, Partial Differential Equations

**Stanford University, Summer Session** Jun–Aug 2022  
*Data Structures and Astrophysics; 8 credits; GPA: 3.8/4.0*  
*Palo Alto, CA*

## RESEARCH & PROFESSIONAL EXPERIENCE

**NYU High Speed Research Network Lab** May 2026–Present  
*Network Programmer*  
*New York, NY*

- Built an edge-offloaded RGB-D SLAM pipeline with ROS 2, RTAB-Map, Docker, and Kubernetes, moving mapping workloads from a Raspberry Pi robot to remote workers.
- Developed a ROS 2–Corelink bridge for cross-network message transport; implemented serialization, fragmentation, reassembly, and UDP pacing for high-throughput RGB-D streams.
- Patched TCP stream reassembly and UDP receive-buffer handling in the Corelink C++ client to resolve fragmented-header and datagram-corruption failures during sustained image streaming.
- Evaluated local and offloaded SLAM under controlled latency, jitter, and packet loss, measuring trajectory error (ATE/RPE), frame age, throughput, compute utilization, and energy use.

**Shanghai Data Exchange** Jun–Aug 2024  
*IT Operations Intern*  
*Shanghai, China*

- Automated scheduled security-audit reporting with Selenium and Baidu OCR, from authentication and screenshot capture to text extraction and report generation.
- Deployed SmokePing probes across approximately 40 hosts and built a Flask/MySQL dashboard to monitor server TCP processes for incident triage.

## ENGINEERING PROJECTS & LEADERSHIP

**NYU Robotics Club** Feb 2026–Present  
*Data Center Robot Software Lead*  
*Brooklyn, NY*

- Led software integration for a ROS 2 mobile robot for remote data-center inspection, combining teleoperation, robot telemetry, and live video feedback.
- Built a video pipeline with the Viam C++ SDK and OpenCV; benchmarked end-to-end latency across resolutions and frame rates to identify encoding and transport bottlenecks.

**NYU RoboSub** Jan 2024–Present  
*Motion Team Lead*  
*Brooklyn, NY*

- Developed waypoint and path-following logic for underwater missions, integrating YOLO detections, task state machines, and closed-loop control for gate traversal, path tracking, and torpedo deployment.
- Implemented and tuned PID motion controllers using IMU and depth feedback; deployed the control stack on NVIDIA Jetson with roserial communication to embedded controllers.
- Ported the vehicle stack to Gazebo 11 and ROS Noetic with Project DAVE hydrodynamics, buoyancy, and thruster models; authored the URDF and competition worlds for repeatable autonomy tests.

**OSIRIS Lab** Oct 2024–May 2025  
*Researcher*  
*Brooklyn, NY*

- Designed a power-analysis platform for an STM32 AES-128 target; implemented SPI acquisition and I<sup>2</sup>C trigger synchronization, then applied correlation power analysis to recover key-byte candidates.
- Completed the OSIRIS Recruit CTF and participated in weekly security scrimmages covering binary exploitation, reverse engineering, cryptography, and web security.

**Sysmon** · Rust, Tauri, React, TypeScript

- Built a cross-platform system monitor with typed IPC commands for CPU, memory, disk I/O, and network metrics; synchronized shared system state with a Rust Mutex and serialized data with serde.

## TECHNICAL SKILLS

**Languages:** C, C++, Python, Bash, Rust, TypeScript, SQL  
**Robotics & Vision:** ROS 2, ROS Noetic, RTAB-Map, ORB-SLAM3, Nav2, Gazebo, YOLO, OpenCV, MATLAB  
**Systems & Infrastructure:** Linux (Ubuntu/Fedora), TCP/UDP, Corelink, Zenoh, Docker, Kubernetes, Git  
**Web & Data:** React, Tauri, Flask, Rails, MySQL